

BT5110 Data Management and Warehousing

AY 2026/27 Semester 1

Group Project

By: Group 5

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1. Entity-Relationship Diagram

Figure 1's entity–relationship diagram sets out the conceptual schema for the Tour de France database, showing how riders, teams, stages, and results connect. It mirrors the race's real structure and rules while staying logically sound and free of redundancy.

Each entity represents a real component of the competition — countries, teams, riders, locations, stages, results, exit, and classification awards — linked through cardinalities and participation constraints that enforce realistic rules: a rider belongs to one team, a team is tied to one country, and every stage has a defined start and finish.

In the diagram, foreign keys are left out, since they're a relational-schema concept that only appears once the model is translated into tables. This design provides the foundation for that translation into a normalized relational schema, supporting rider performance, team standings, and race outcomes while preserving referential integrity for later analysis.

1.1 Entities and Relationships

Table 1. Core Entities

Entity	**Description**	**Key Attribute**	**Other Attributes**
Country	A nation participating via its teams and/or riders. Recorded even if it currently has no team or rider registered.	country_code	country_name
Team	A professional cycling team competing in the race. Must have at least one rider to be recorded.	team_name	— (country and riders captured via relationships, not attributes)
Rider	An individual cyclist competing for a team. Belongs to exactly one team and represents exactly one country, independently of each other.	bib_no	rider_name, dob, is_leader
Location	A place used as the start or finish point of a stage. Only locations actually used in that role are stored.	location_name	— (country captured via relationship)

Stage	A single day's leg of the race, run between a start and a finish location.	stage_id	stage_date, distance_km, stage_type
Result (associative entity)	A rider's outcome for a specific stage, identified jointly by Rider and Stage via completes.	(inherited: bib_no + stage_id)	rank, finish_time, bonus, time_penalty, adjusted_time (derived)
Exit (associative entity)	A rider's withdrawal from the race, identified jointly by Rider and Stage via *withdraws* — the stage being the one the rider withdrew at.	(inherited: bib_no + stage_id)	exit_reason
Classification/ Award	A competition award (General Classification, Points, King of the Mountains, Young Rider, Best Team).	award_type	is_team

Table 2. Key Relationships and Constraints

Relationships	**Participating Entities & Cardinality**	**Description**
represents	Country (0,N) — Team (1,1)	Each team represents exactly one country; a country may be represented by zero or more teams.
based in	Country (0,N) — Rider (1,1)	Each rider is based in exactly one country; a country may have zero or more riders based in it.
consist of	Team (1,N) — Rider (1,1)	Each rider belongs to exactly one team; each team consists of one or more riders.
hosts	Country (0,N) — Location (1,1)	Each location belongs to exactly one country; a country may host zero or more locations.
starts	Location (0,N) — Stage (1,1)	Each stage starts at exactly one location; a location may be the start of zero or more stages.

ends	Location (0,N) — Stage (1,1)	Each stage ends at exactly one location; a location may be the finish of zero or more stages.
completes	Rider (0,N) — Stage (0,N) — Result (1,1)	Rider and Stage jointly identify a Result row; a rider completes many stages, a stage has results from many riders.
withdraws	Rider (0,1) — Stage (0,N) — Exit (1,1)	Rider and Stage jointly identify an Exit row; a rider withdraws at most once, at one particular stage.
wins	Team (0,N) — Rider (0,N) — Classification/Award (1,N)	A single relationship linking Team, Rider, and Award: an award is won by a rider or a team, and a rider or team may win zero or more awards.

1.2 Design Rationale and Assumptions

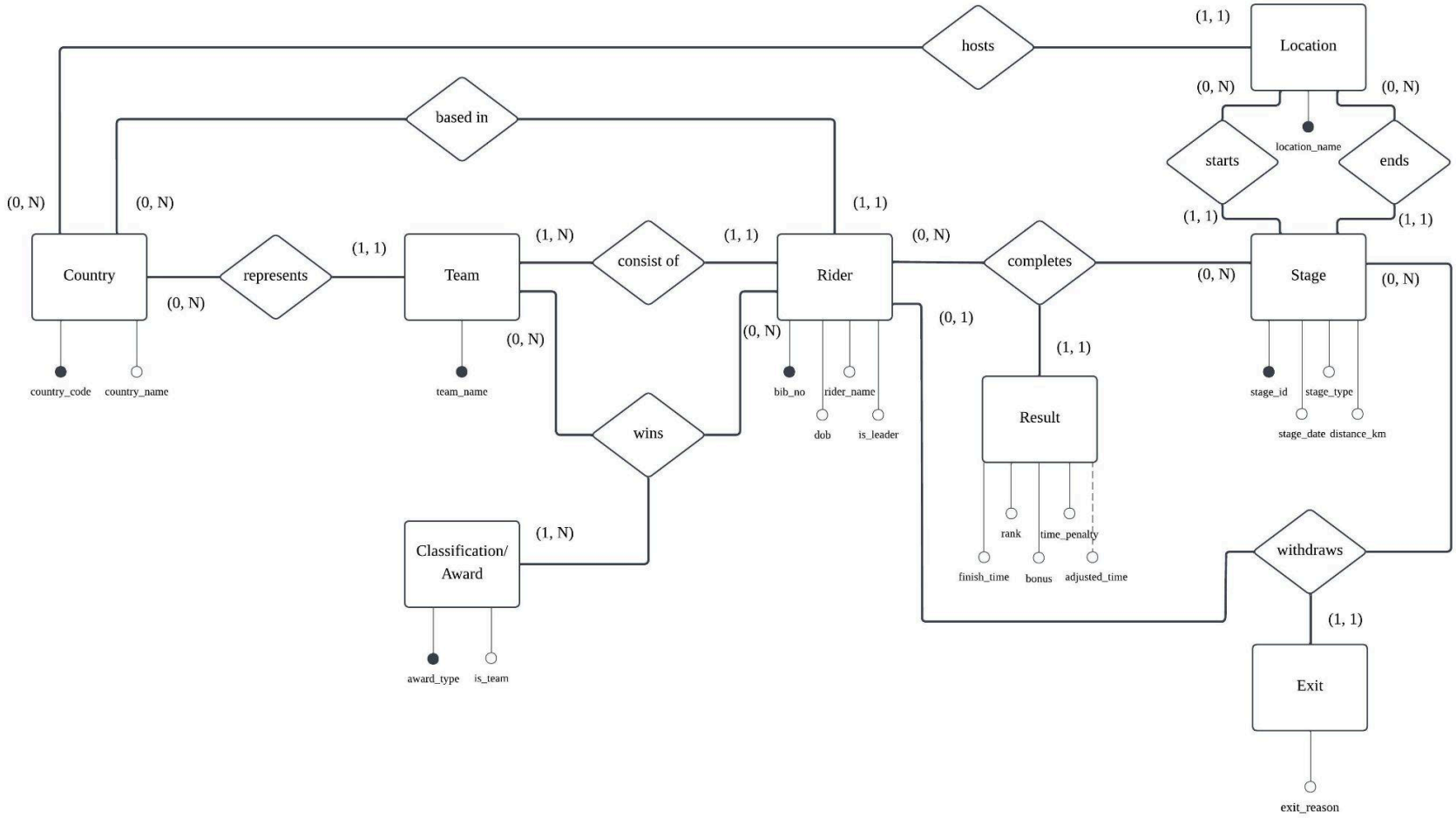
The rule that a location is stored only if it starts or finishes at least one stage can't be captured directly in the diagram. It's a disjunctive condition — a location is valid once it appears in *starts* or *ends*, not necessarily both — and each relationship's participation constraint only limits that relationship on its own, so a location tied to neither one still satisfies both taken individually. That kind of either/or condition spanning two separate relationships sits outside what a conceptual ER diagram can express, so it's left documented here as an assumption rather than built into the model.

Result and Exit share the same two-owner structure. Both are drawn as double-bordered associative entities identified jointly by Rider and Stage — Result through *completes*, for stages a rider actually finished, and Exit through *withdraws*, for the stage a rider withdrew at. Neither has a key of its own; each borrows its identity from the pairing of its two owners, which turns out to be useful beyond just modelling convenience. Because both facts now carry an explicit stage_id, the rule that no results should exist for a rider past their exit stage becomes a straightforward comparison rather than something that has to be inferred — the exit's stage_id simply marks the cutoff for that rider's Result rows.

Team leadership, by contrast, didn't need a relationship of its own. is_leader sits directly on Rider instead of being expressed through a separate led-by relationship — since every rider already belongs to exactly one team via *consist of*, flagging is_leader on Rider is enough to guarantee a team's leader is drawn from its own riders.

Award exclusivity works the same way, through a flag rather than extra structure. Classification/Award carries an is_team attribute alongside award_type, distinguishing a team-level award (Best Team) from the four rider-level awards — General Classification, Points, King of the Mountains, and Young Rider — that all feed into the single wins relationship.

Figure 1:


hosts
(1, 1)
Location
(0, N)
(0, N)
based in
location_name
starts
ends
(0, N)
(0, N)
(1, 1)
(1, 1)
(1, 1)
Country
represents
(1, 1)
Team
(1, N)
consist of
(1, 1)
Rider
(0, N)
completes
(0, N)
Stage
(0, N)
(0, N)
(0, N)
(0, 1)
(1, 1)
country_code
country_name
team_name
bib_no
rider_name
dob
is_leader
Result
stage_id
stage_type
stage_date
distance_km
wins
rank
time_penalty
finish_time
bonus
adjusted_time
withdraws
(1, N)
Classification/
Award
(1, 1)
Exit
award_type
is_team
exit_reason